

Machine-Learning & DFT Guided Electrode Design for Oxide-based Supercapacitors: From Data Mining to Experimental Validation

(This project report has been submitted in partial fulfillment of
the requirements for the degree of Bachelor of Science in
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SUBMITTED BY:

Exam Roll No.: 232318

Reg. No.: 20217111484

Session: 21-22

Exam Roll No.: 232339

Reg. No.: 2021911464

Session: 21-22

**DEPT. OF ELECTRICAL AND ELECTRONIC ENGINEERING
UNIVERSITY OF DHAKA**

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ABSTRACT

The conventional trial and error based approach of designing supercapacitor electrodes is slow, costly and intuition driven. Simple perovskite oxides (ABO_3) and Double Halide perovskite oxides ($\text{A}_2\text{BB}'\text{O}_6$) have emerged as a promising class of materials for pseudocapacitors. However, their compositional design space remains largely uncharted. It is one of the main barriers of developing next generation of electrodes. We employ machine learning as a screening tool to pinpoint the features which has the greatest influence on Specific Capacitance of perovskite oxide. The first phase of the work consists of a literature-based materials survey, the construction of a descriptor dataset and outline of a machine learning based pre-screening framework combining candidate predictive models with feature ranking to identify key properties. The dataset is composed of features like composition, synthesis conditions, structural properties and electrochemical properties. On the basis of the machine learning screening results, the next phase focuses on the targeted synthesis of candidate compositions which are identified as most promising. These materials will be characterized using X-ray diffraction, Braunauer-Emmett-Teller analysis, Raman and X-ray Photoelectron Spectroscopy to establish a comprehensive correlation between structure and property. This submission presents only the research planning, methodology and literature review related to the proposed approach. No experimental or computational results are reported at this stage. The actual significance of this work lies in establishing a pre-screening pipeline which is data-driven and capable of narrowing the perovskite candidate space before actual synthesis work. The pipeline hence reduces the experimental effort and accelerates the discovery of high specific capacitance electrode materials for future supercapacitors.

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CHAPTER 1

INTRODUCTION

1.1 Background and Context

Supercapacitors bridge conventional capacitors and batteries, offering higher power density and cycle life than batteries and greater energy density than capacitors. The primary usage areas of supercapacitors are Hybrid Electric-vehicles (HEV), Electric-vehicles (EV), Fuel Cell Vehicles (FCV), alongside several consumer and enterprise electronics spaces [1], [3], [16]. They store charge by electric double-layer capacitance (EDLC) at high specific-surface-area carbons and by pseudo capacitance from reversible redox processes on transition-metal-oxides (TMO) and conducting polymers [1], [2], [3], [6]. TMOs are especially attractive for their multiple oxidation states, high specific capacitance, and stability [4].

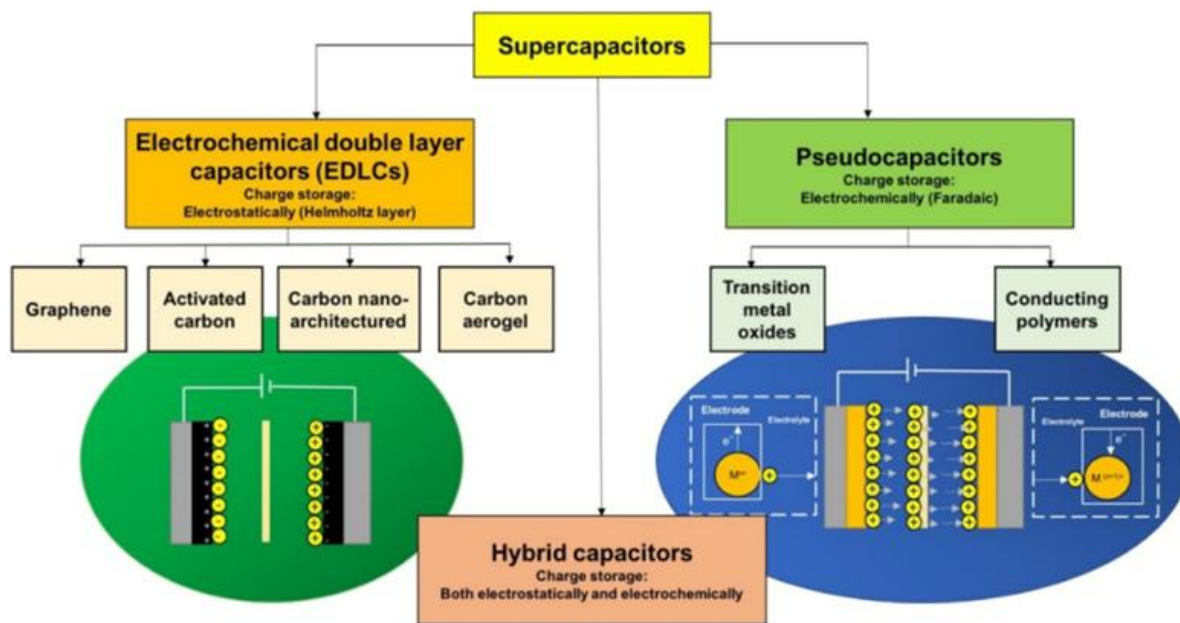


Fig. 1.1: Charge-storage mechanisms in supercapacitors: Electric Double-Layer Capacitance, Pseudo capacitance and Hybrid [1], [4].

1.1.1 Material of Choice: Perovskite Oxides

Perovskite oxides are a highly attractive class of materials for the supercapacitor electrode design, due to their unique combination of properties aiming to address the limitations of traditional electrode materials.

Standard perovskite-oxide structures (ABO_3) place large A-site cations (e.g. Sr, La, Gd) at the cube corners and smaller B-site transition metals (Fe, Ni, Mn, Co, Cr,) within corner-sharing BO_6 octahedra [3], [11], [14] (refer to Fig. 1.2(a),(b)). Their tolerance for cations of differing radius and valence gives wide compositional tunability [3], [11]. A-/B-site substitution/doping and oxygen-vacancy engineering govern conductivity, while the anion-intercalation charge storage governs capacitance [3], [14]. Structural variants to the perovskite-oxides: double-perovskites ($\text{A}_2\text{BB}'\text{O}_6$) with two alternating B-site cations; and layered Ruddlesden–Popper phases ($\text{A}_{n+1}\text{B}_n\text{O}_{3n+1}$).

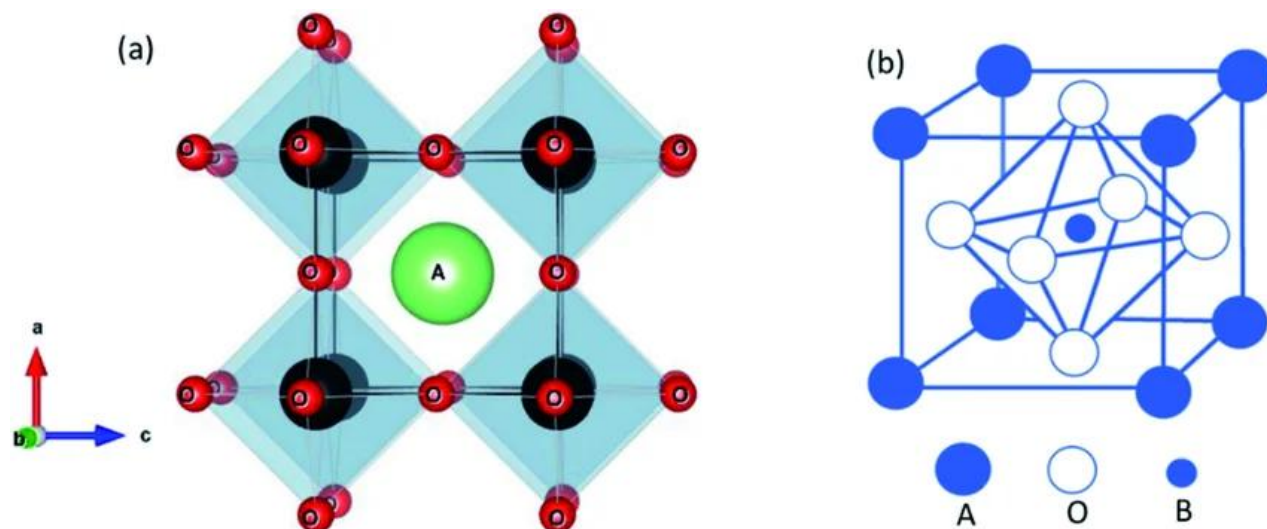


Fig. 1.2: ABO_3 perovskite structure. (a) Polyhedral view; corner-sharing BO_6 octahedra with A-site cation occupying the central 12-coordinate cavity and oxygen anions at octahedral vertices. (b) Unit-cell view; A-site cations at cube corners; B-site cation at body centre; oxygen anions at face centres [3].

1.1.2 Computation-guided Closed-loop Pipeline

Specific capacitance (C_{sp}) of supercapacitors is dependent on several non-linear variables, namely structure, composition, specific surface area, morphology, doping, and synthesis processes, that make pure experimental trial-error-based optimization difficult and expensive [3], [4], [14]. This raises the need for a descriptor-based machine-learning (ML) approach that learns structure–processing–performance relationship from existing literature [6], [11]. Existing ML studies, however, often omit stability modelling, consider a single-material system, pool heterogeneous chemistries into blackbox models, and rarely validate predictions against first-principles calculations and targeted synthesis [6], [11], [15].

1.2 Objectives

The primary target for this project is to develop a perovskite-oxide specific ML framework that links data-driven prediction to first-principles Density Functional Theory (DFT) and experimental validation. This requires overcoming several hurdles: curating the composition–structure–synthesis dataset from inconsistent literature data; ensuring the predicted candidates' physical synthesizability and effective nanoscale synthesis; approached through ML screening, DFT validation, and targeted experiments.

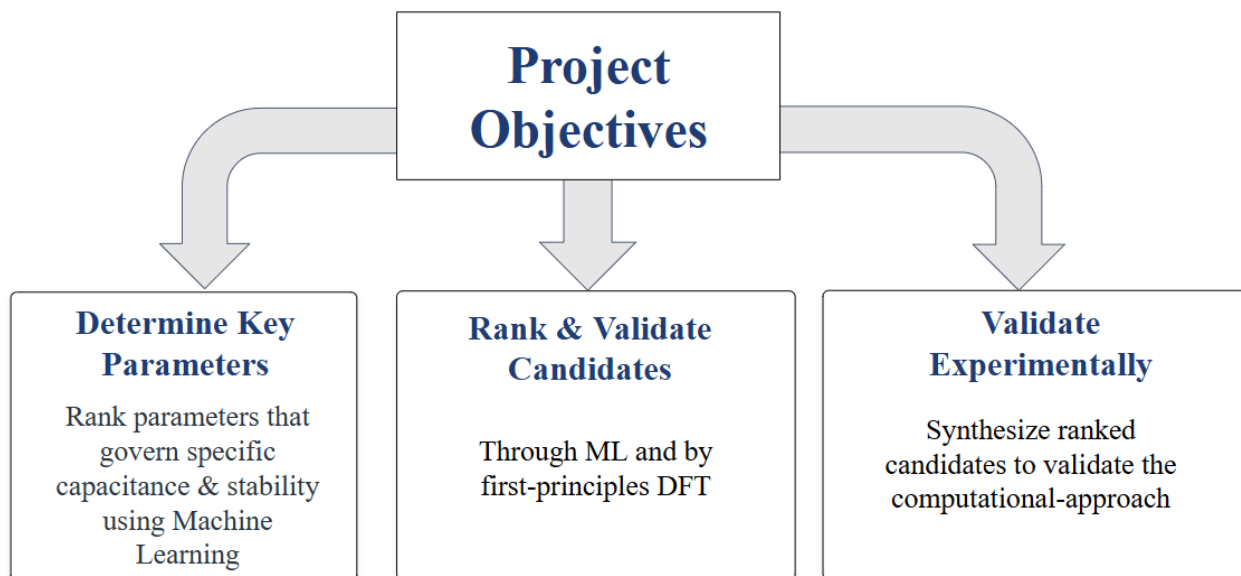


Fig. 1.3: The outline of the Objectives of our project.

So, the main objectives list come down to:

1. Determining key parameters that strongly govern C_{sp} and stability in perovskite-oxide electrodes, utilizing explainable AI (XAI).
2. Producing ranked perovskite composition candidates, predictively with high specific capacitance and stability; validated through by both first-principles DFT and experiment.
3. Establishing a closed-loop design-to-synthesis (from computational modelling to experimental feedback) pipeline that cuts trial-and-error optimization in perovskite-oxide electrode material discovery

Chapter 2

LITERATURE REVIEW

Summary of this Chapter :

This chapter reviews the existing literature on oxide-based supercapacitor electrode materials and prediction of C_{sp} using ML. It covers spinel oxides, layered/nanostructured oxides, carbon-based ML benchmarking studies, and mixed-material ML frameworks, followed by ML-explainability studies relevant to feature interpretation. The chapter also puts the gaps across these studies together and proposes a descriptor framework.

Chapter Outlines:

- 2.1 Background and Material Importance
- 2.2 Machine Learning on Spinel Oxide Electrodes
- 2.3 Layered and Nanostructured Oxide Electrodes
- 2.4 Carbon-Based ML Benchmark Studies
- 2.5 Mixed-Material ML Benchmarking Frameworks
- 2.6 ML with Explainability (XAI) in Electrode Material Studies
- 2.7 Synthesized Literature Gaps and Proposed Novelty
- 2.8 Final Project Proposal

2.1 Background and Material Importance

Conventional capacitors have low specific energy and conventional batteries have low specific power. In fig2.1, we can see this very gap is bridged by Supercapacitor. They offer high power density, fast charge-discharge kinetics, and long cycle life [1], [2], [3].

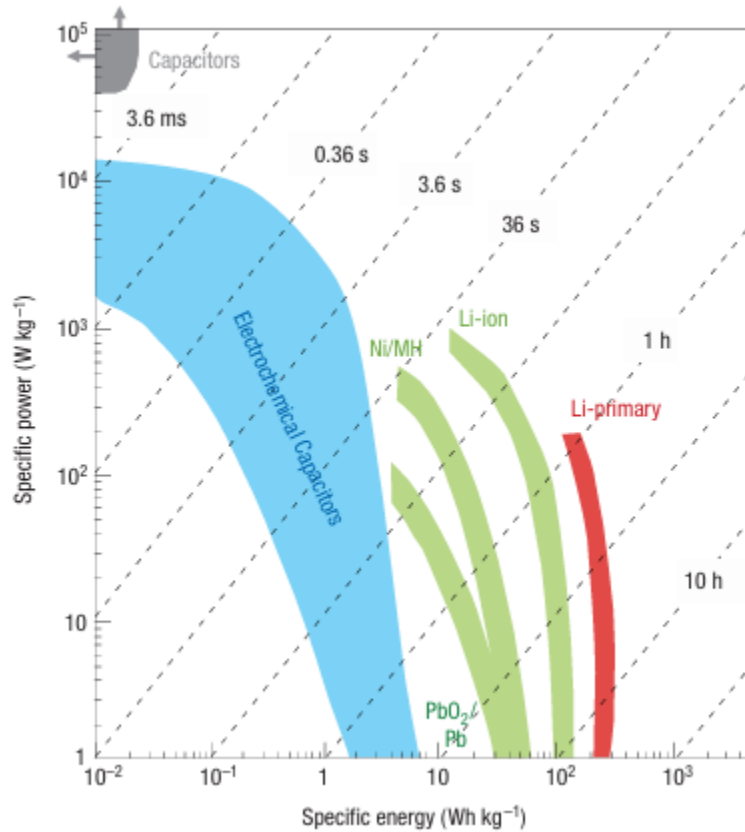


Fig. 2.1: Specific power against specific energy for various electrical energy storage devices [1].

Based on the charge-storage mechanism, supercapacitors are broadly classified into electric double-layer capacitors (EDLCs) and pseudocapacitors. EDLCs store charge electrostatically at the electrode-electrolyte interface (typically carbon-based materials), whereas pseudocapacitors do the same using Faradaic redox reactions at the electrode surface [3], [4].

Table 2.1: The three different Charge-storing methods

Charge-storage Mechanisms	Description
EDLC	Electrostatic storage; high-surface-area carbon electrodes
Pseudo-capacitance	Reversible Faradaic redox; TMOs & conducting polymers
Hybrid	Combines both

Transition-metal oxides have been extensively proposed in the literature as alternatives to carbon-based EDLC electrodes since they have multiple oxidation states, higher theoretical specific capacitance, and tunable electronic structure [3], [4]. However, the electrochemical performance of these oxide electrodes is strongly governed by a combination of factors including synthesis route, crystal phase purity, surface area, particle morphology, and doping . [4], [5]. Fig 2.2 depicts the dependence of such factors on Specific capacitance.

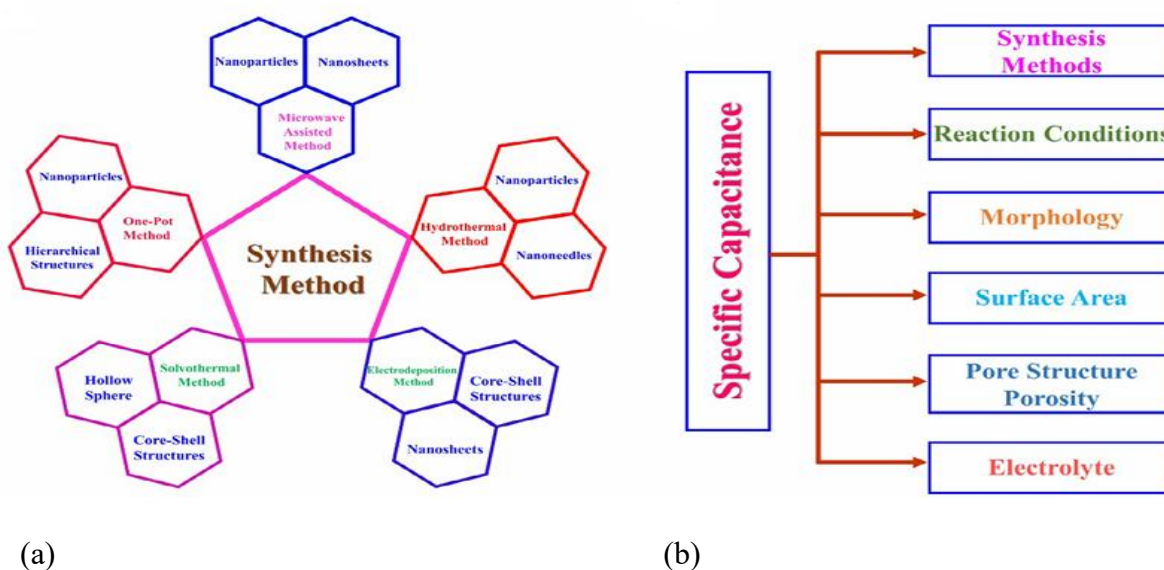


Fig. 2.2 (a) Correlation between synthesis method and observed morphology. (b) Parameters influencing specific capacitance [13].

This multi-parameter dependency makes the whole process trial and error based, causing loss of time, money and resources. An ML approach that can learn the structure–synthesis–property relationship directly from existing experimental data and guide the selection of new candidate materials before synthesis solves the problem and brings forward a much more effective pipeline [5], [6].

2.2 Machine Learning on Spinel Oxide Electrodes

AB₂O₄- Spinel oxides are among the most widely used pseudocapacitive oxide families. Their mixed-valence cation sites and good electronic conductivity make them one of the best candidates. In Fig. 2.3, we can observe Tang et al. applied and compared nine different ML models to predict the C_{sp} of MgCo₂O₄ electrodes from structural and electrochemical descriptors [7]. Their study is a strong methodological reference for the present project because it demonstrates rigorous model benchmarking practice, including the use of a Williams plot and applicability-domain (AD) analysis to identify outliers and confirm that model predictions remain within a statistically reliable data range [7].

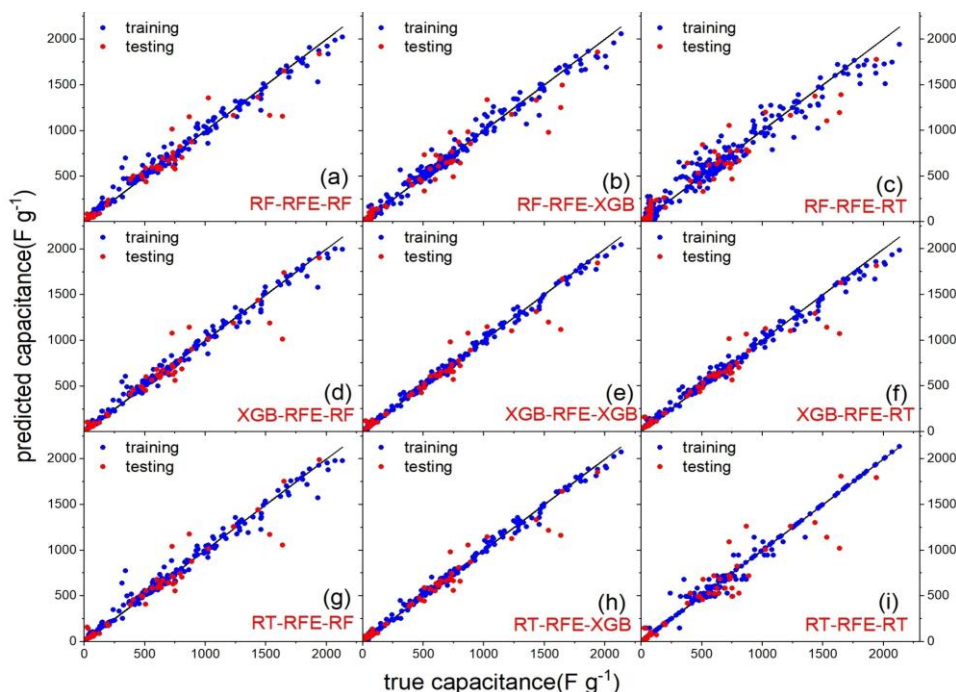


Fig. 2.3 Predicted vs. true specific capacitance for MgCo₂O₄ using nine RFE-selector × regressor model combinations (RF, XGBoost, RT) [7] .

However, the scope of this study is limited to a single material class (MgCo_2O_4) and the feature set is restricted to structural and electrochemical descriptors. Other parameters of the synthesis process such as calcination temperature, calcination time, and synthesis method are not provided as ML inputs [7]. This leaves an open question of whether process level descriptors, which are known to strongly influence phase purity and morphology, would improve prediction accuracy or not.

2.3 Layered and Nanostructured Oxide Electrodes

Nanostructured transition metal oxides have also attracted considerable attention as supercapacitor electrodes because of their high surface area, short ion diffusion pathway and abundant electrochemically active sites which enhance charge storage. Vasistha et al. developed a combined experimental and ML framework to investigate MnO_2 nanorod-based supercapacitor electrodes integrated with graphene oxide nano-rings (GONs). The study also involved the synthesis and structural characterization of the nanostructured electrode, followed by electrochemical evaluation and the development of ML models capable of predicting specific capacitance from experimentally derived features. Fig. 2.4 shows the results of the characterization performed in this literature. [8].

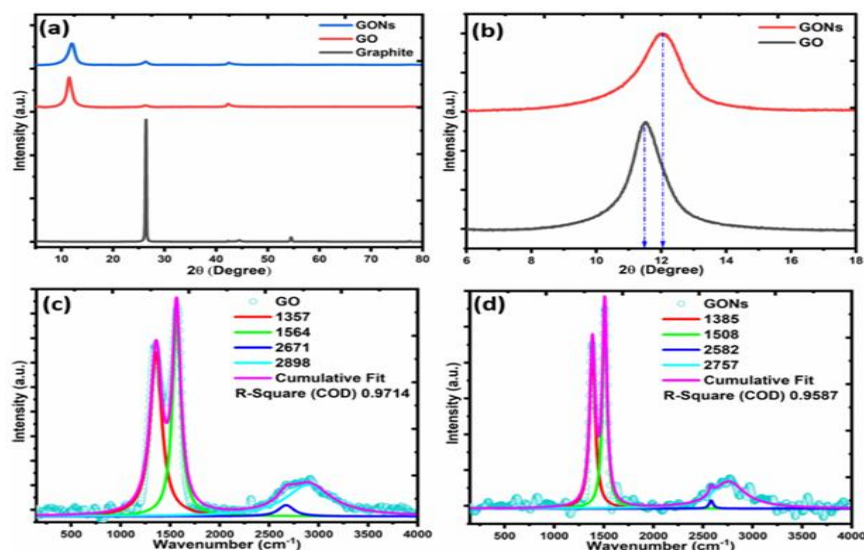


Fig. 2.4 (a) XRD spectra, (b) magnified view of the (001) peak, and (c)–(d) Raman spectra of GO and GONs [8].

While this work provides valuable insight into how physical morphology descriptors relate to electrochemical performance, its scope is narrow since it's restricted to a single morphology (nanorods) within a single material system (MnO_2). The work is limiting the generalizability of its conclusions to other oxide families such as perovskites or double perovskites [8].

2.4 Carbon-Based ML Benchmark Studies

Carbon-based ML studies remain valuable as methodological templates though the current project focuses on Oxide-based material. Shrivastava and Dubey built an ML pipeline for activated carbon electrodes using six descriptors : specific surface area (SSA), pore size (PS), pore volume (PV), pore width (PW), nitrogen content (NC) and Raman I_D/I_G ratio. They also established a complete workflow encompassing data gathering, feature engineering, model selection, and evaluation shown in Fig. 2.5 [9].

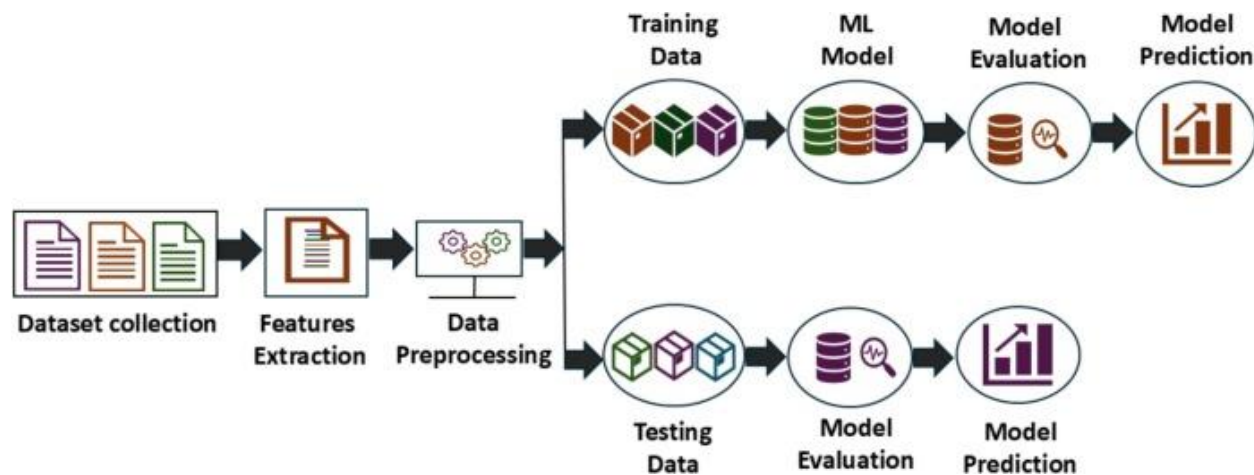


Fig. 2.5 ML Pipeline for activated Carbon electrodes using six descriptors [9].

This pipeline is a perfect methodological template for the present project's ML pipeline design. However, it is not oxide-based, inconsistent to oxide-material study and synthesis conditions are not included as descriptors. Hence, the framework does not capture how the electrode was made, only discusses its structure [9].

2.5 Mixed-Material ML Benchmarking Frameworks

A more directly relevant work is a benchmarking study on aqueous symmetric supercapacitors by S Ghosh. Their work compared support vector regression (SVR), random forest (RF), neural-network classifier (NNC) and multilayer perceptron (MLP) combined with principal component analysis (PCA) across eleven features [10]. Moreover, this study organized its features into three descriptor categories: synthesis/thermal, structural, and electrochemical as shown in Fig.

2.6[10]. This very grouping structure is directly adaptable to the present project's descriptor framework

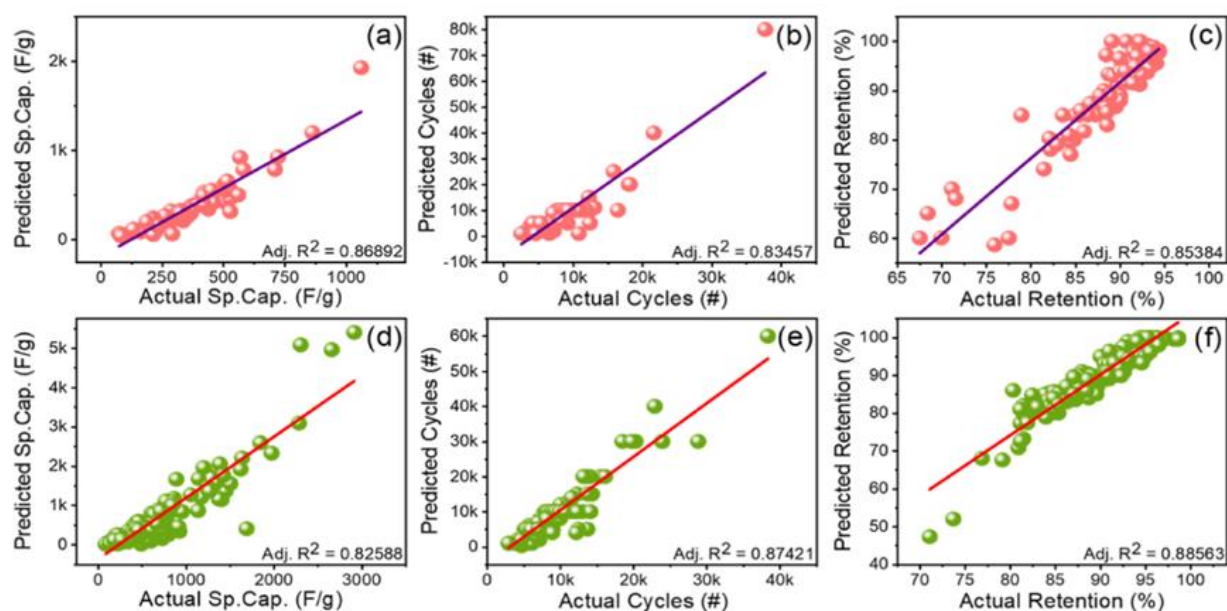


Fig 2.6: Output performance features prediction of MXene-based SCs with the help of the RF algorithm (GCD and CV datasets). Prediction of (a) specific capacitance, (b) cyclic stability, and (c) capacitive retention based on the GCD dataset. Prediction of (a) specific capacitance, (b) cyclic stability, and (c) capacitive retention based on CV dataset. The data are fitted with a linear fitting method. [10].

Another notable strength of this work is that the proposed machine learning framework was not limited to computational benchmarking but also validated through experimental fabrication and testing. Besides, in Fig. 2.7, we can observe the results of electrochemical performance evaluation of the electrodes. The fabricated electrodes exhibited specific capacitance values that

closely matched the ML predictions. It confirmed the reliability of the proposed framework for guiding material selection and electrode design in future. This combination of predictive modeling with experimental verification enhances the credibility of the developed models and demonstrates the practical applicability of ML for accelerating supercapacitor material discovery and optimization.[10]

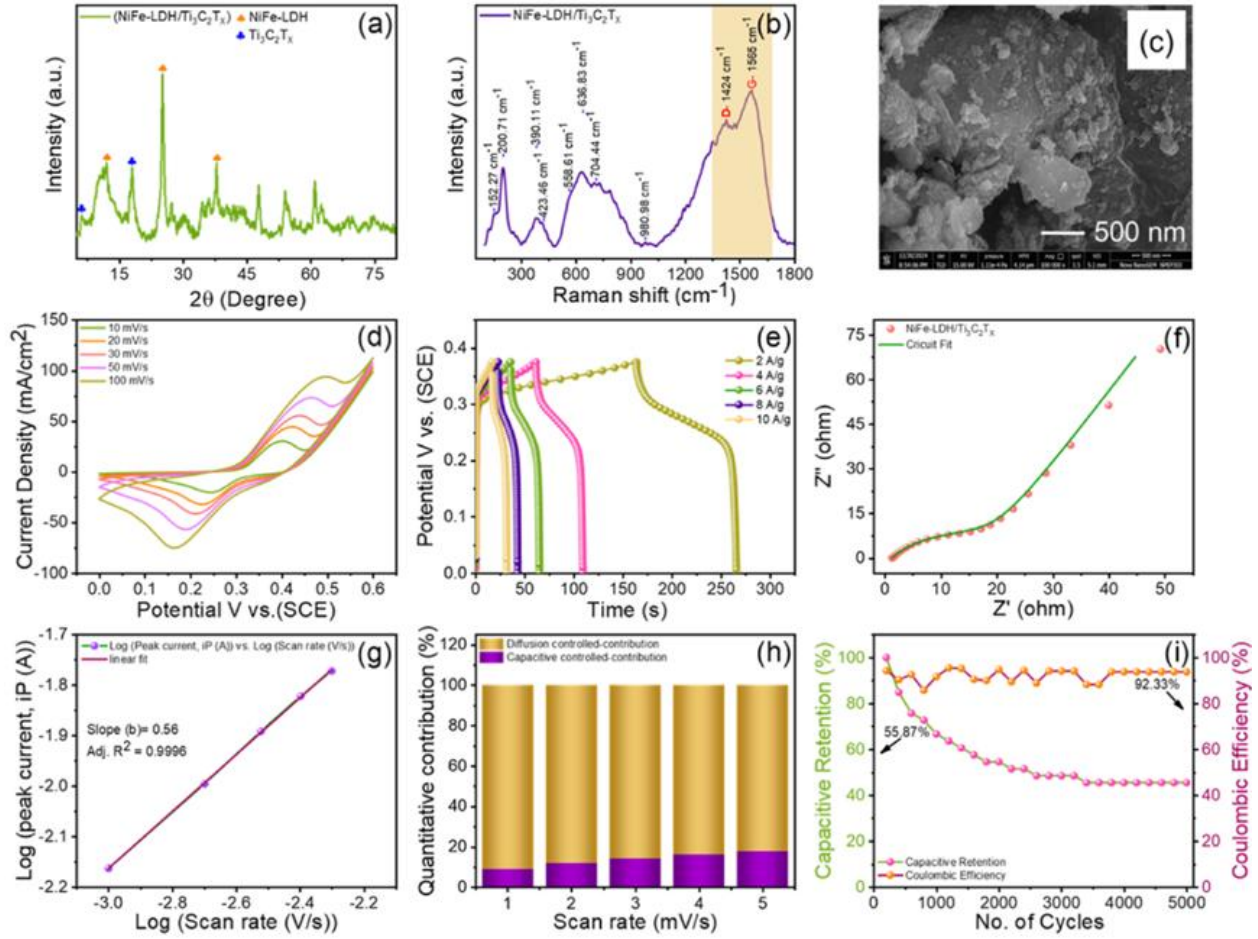


Fig. 2.7: Experimental validation of ML results by fabricating an MXene-based SC based on the GCD dataset. (a) XRD, (b) Raman, and (c) SEM results of the synthesized NiFe-LDH/MXene material. (d) CV, (e) GCD, (f) EIS results of the NiFe-LDH/MXene SC. (g–h) Capacitive and diffusion-driven contribution of the NiFe-LDH/MXene SC at various scan rates (1–5 mV/s) in 1M KOH electrolyte, (i) Capacitive Retention and Coulombic efficiency of NiFe-LDH/MXene SC [10].

However, the reported R^2 values for this study were modest, ranging from approximately 0.35 to 0.41 across different models [10]. This relatively low predictive performance suggests that the limiting factor was likely feature selection quality or dataset scale rather than model architecture. This is an important signal that careful, domain-informed descriptor design is at least as important as model choice when building a reliable Csp-prediction pipeline.

2.6 ML with Explainability (XAI) in Electrode Material Studies

XAI techniques have started to be applied in electrode material studies. A recent ML and XAI-focused paper used a Pearson correlation coefficient (PCC) heatmap combined with feature reduction techniques resulting in improvement of model interpretability of capacitance predictions [11]. This provides a relevant methodological precedent for the SHAP (SHapley Additive exPlanations) analysis planned for the present project (fig 2.8).

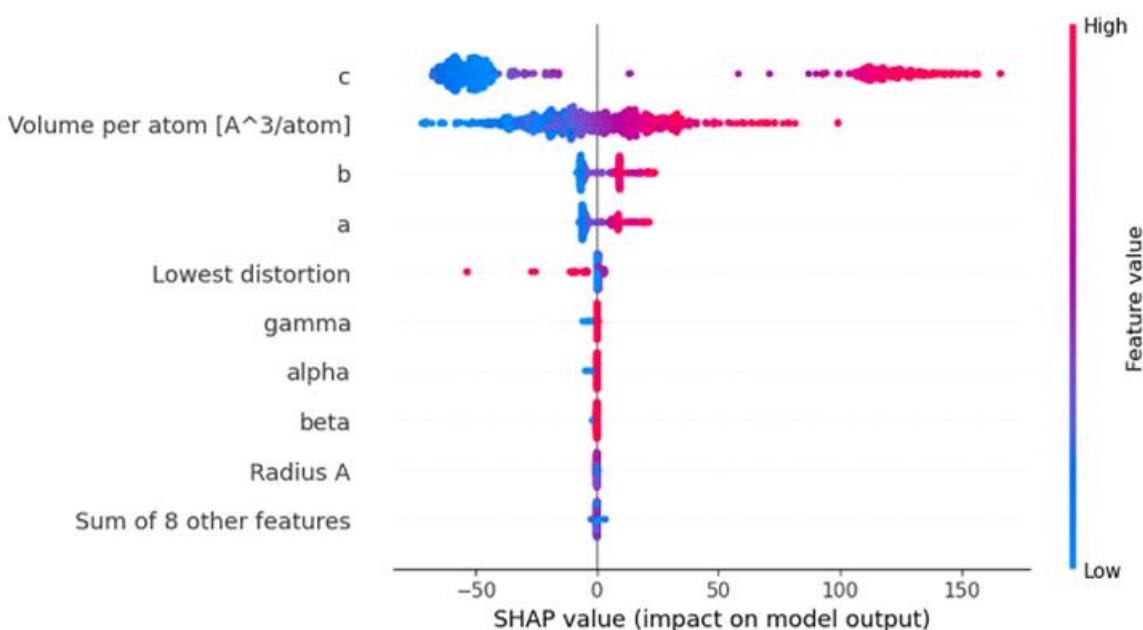


Fig. 2.8: SHAP analysis of specific capacitance prediction showing key feature importance. [11]

Nevertheless, this study is limited by a small dataset size and a focus on a single dominant material family. Its interpretability analysis stops at explaining the mode. It is not extended to guide any follow-up, targeted experimental synthesis creating a wide research gap [11].

2.7 Synthesized Literature Gaps and Proposed Work

Collectively, the literature reviewed above reveals four consistent gaps:

1. No unified descriptor dataset across perovskite families:

None of the existing study builds a unified descriptor dataset consisting of both simple perovskite (ABO_3) and double perovskite ($\text{A}_2\text{BB}'\text{O}_6$) oxides. Most studies are confined to a single material class or morphology [7], [8], [10].

2. Synthesis-process parameters are underused as ML features:

Across the spinel oxide [7], carbon-based [9], and mixed-material [10] studies, synthesis/process parameters, calcination temperature, calcination time, synthesis method, and doping ratio are inconsistently or rarely incorporated as ML inputs. The most common parameters that are used are structural and electrochemical descriptors, even though the other parameters are known to directly govern phase purity, morphology, and defect density which directly and indirectly affect Specific Capacitance.

3. No feedback loop from ML screening to experimental validation.

None of the reviewed studies close the loop from ML-based candidate screening to targeted synthesis and experimental confirmation of the predicted material. ML is used purely as a predictive tool on existing datasets rather than a prospective design tool [7], [8], [10], [11].

4. Explainability is underused as a design bridge:

Feature-importance/explainability methods such as SHAP/PDP/ICE remain under-used as a bridge connecting computational screening to actual experimental decision-making. Even if interpretability is attempted (e.g., PCC heatmaps [11]), it is not carried forward into guiding new synthesis choices hence creating a research gap.

2.8 Final Project Proposal

This work proposes to address the gaps in existing literature through:

1. Developing a unified descriptor framework for ABO_3 and $\text{A}_2\text{BB}'\text{O}_6$ perovskite oxides.
2. Integrating synthesis-process, structural, and electrochemical descriptors into ML models.

3. Using explainable-AI (XAI) explainability methods to identify influential material descriptors.
4. Validate ML-selected candidate materials, through first-principles DFT and experimentally, establishing a complete Computation-guided optimisation workflow.

CHAPTER 3

PROPOSED METHODOLOGY

Chapter Summary

This chapter presents a two-stage, computation-guided closed-loop methodology for ranking candidate materials for high-performance perovskite oxide supercapacitor electrodes. The chapter details the first-semester's completed work, the dataset-construction methodology, descriptor taxonomy and ML/DFT framework design. It also includes the detailed plan for the second-semester, the equipment required for synthesis and characterization, expected outcomes, and a Gantt chart outlining the timeline.

Chapter Outlines

3.1: Overall Design of the Project

3.2: Work Done During the First Semester

3.3: Detailed Work Plan for the Second Semester

3.4: Expected Outcome

3.5 Gantt Chart

3.1 Overall Design of the Project

This work proposes a two-stage computation-guided closed-loop pipeline:

Stage-1: ML and DFT analysis. Literature data are mined and modelled with ML to rank compositions, dopants, electrolytes, and processing methods by gravimetric C_{sp} , while another parallel model ranks the structural stability of promising perovskite compositions. This ensures shortlisted candidates predictively acquiring both high specific capacitance and stability. The ML-proposed shortlist would then be validated by first-principles DFT, which independently verifies whether the favoured candidates are thermodynamically stable and electronically suitable. Candidates confirmed by DFT are carried forward into stage-2; and until both methods agree, the ML–DFT sequence is refined recursively.

Stage-2: Experimental validation. The ML- and DFT-supported candidates are synthesized and characterized to test whether the computational guidance holds experimentally, closing the loop between prediction and validation.

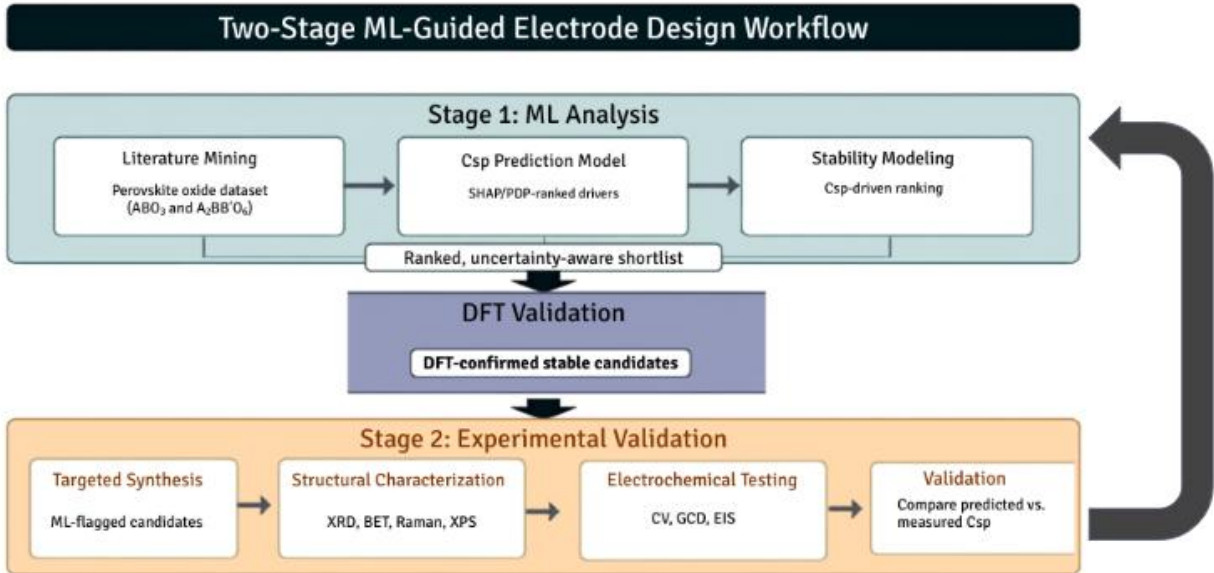


Fig. 3.1: Overall pipeline for ML modelling (C_{sp} and stability), first-principles DFT verification and experimental validation.

3.2 Work Done During the First Semester

The first semester produced the project plan, methodology, and literature-based justification; no experimental or computational results are reported yet. Specifically, the following are achieved:

1. The overall two-stage workflow and its objectives.
2. A literature-based materials survey across perovskite (ABO_3) and related oxide families.
3. Dataset Construction Methodology: Compile from peer-reviewed oxide-electrode literature on a single comparable basis (gravimetric C_{sp} , three-electrode), record source/DOI per entry, and exclude non-comparable measurements, reviews, and cases where the oxide's contribution cannot be isolated. Set the target variable as gravimetric C_{sp} (refer to Fig. 3.2).
4. Candidate Feature Categories: composition/structure (e.g. tolerance factor, octahedral factor), synthesis/processing, physico-chemical properties and measurement conditions. (refer to Fig. 3.3).
5. Overall Machine-Learning framework: Baseline plus candidate models (linear/regularized, tree ensembles, kernel-based methods), source-grouped cross-validation to avoid leakage, metrics (R^2 , RMSE, MAE), and SHAP/PDP interpretation.
6. Modelling strategy for stability (utilising features such as formation energy and band gap) and the inverse-design screening that yields a ranked, uncertainty-aware shortlist for DFT validation.

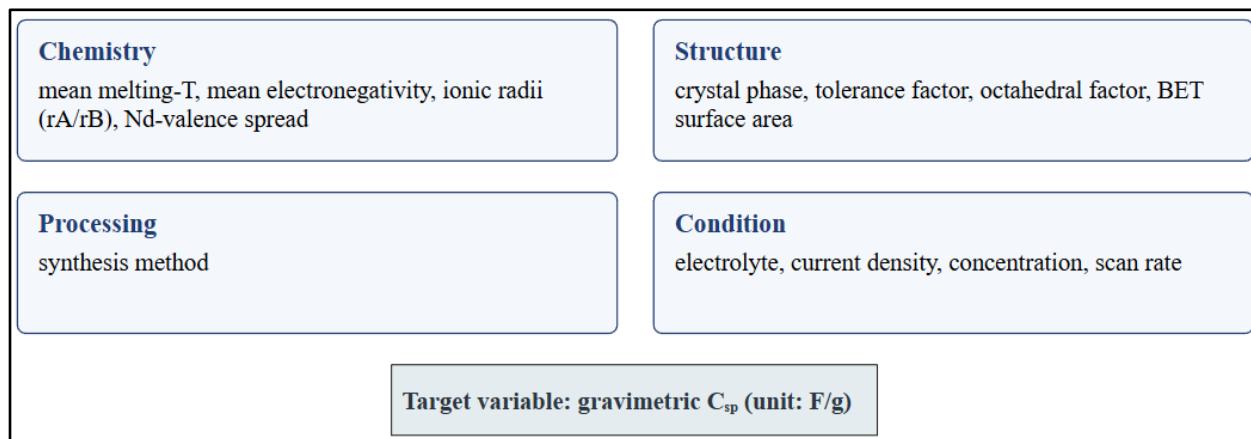


Fig. 3.2: The four descriptor families that are aimed to be considered in the dataset construction approach of the computational part of the project.

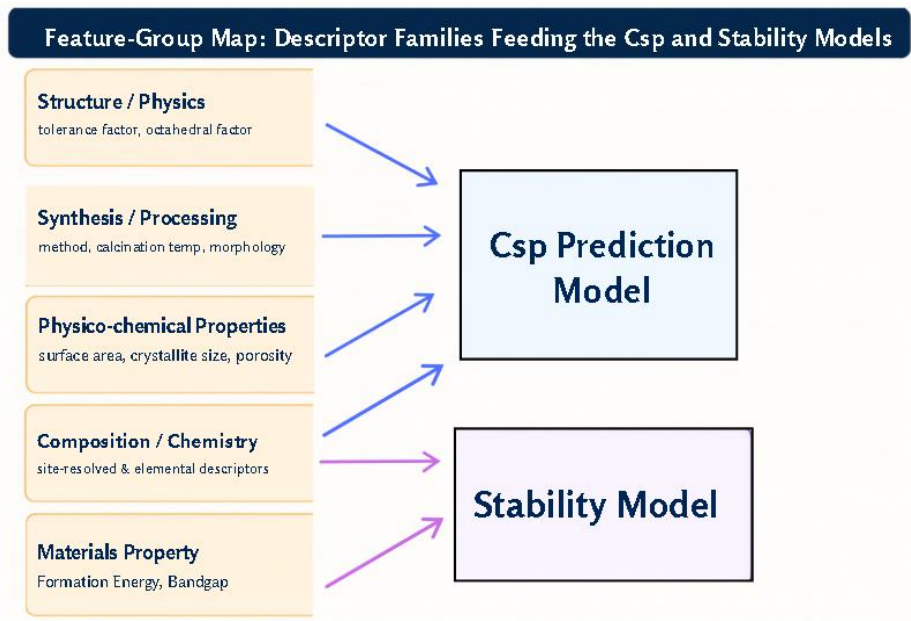


Fig. 3.3: Feature-group map of the descriptor families feeding the C_{sp} and stability models.

3.3 Detailed Work Plan for the Second Semester

Stage-1: ML and DFT Execution

ML-guided ranking:

The dataset will be built, finalized, and cleaned (handling duplicates, outliers, mismatched units, missing values), the feature set will be finalized once the literature-coverage is known. Models will be trained with source-grouped cross-validation and in-fold tuning, the final model selected on results, and SHAP/PDP interpretation used to prioritize the driving chemistries and processing conditions. The stability and C_{sp} models are combined to screen for both performance and stability.

DFT validation:

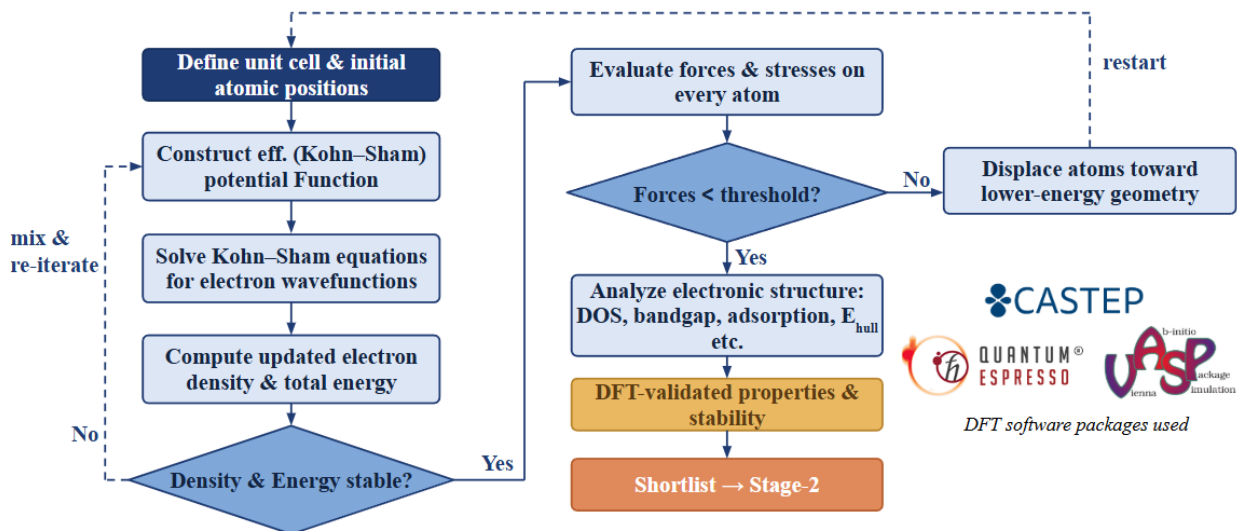


Fig. 3.4: The flowchart showcasing the DFT methodology utilised in the project-work.

The flowchart in Fig. 3.4 shows the whole functionality of first principles DFT in our project work. First and foremost, the DFT workflow starts with gathering .cif files (Crystallographic Information File) that defines the unit cell and initial atomic positions to be input to the system. The output from the system would be the estimation of the properties listed in Table 3.1.

Table 3.1: List of properties to be estimated using Density Functional Theory (DFT)

Property	How It Is Estimated
Band structure / bandgap	Kohn–Sham eigenvalues along k-path
Density of States (DOS)	Electron occupation vs. energy near Fermi level
Formation / cohesive energy	Total energy difference vs. reference states
Adsorption / binding energy	Energy difference before/after adsorbate binding
Work function	Vacuum potential minus Fermi energy (slab calculation)
Charge density distribution	Bader charge analysis on converged electron density
Lattice parameters / elastic constants	Geometry and energy optimization (structural relaxation)
Phonon dispersion / stability	Density Functional Perturbation Theory (DFPT)

Stage-2: Experimental Execution

Addressing Nano-scale synthesis challenge. As high C_{sp} requires nano-engineered electrodes with short ion-diffusion paths and large surface area rather than bulk powders, the target essentially comes down to sub-200 nm particles. Since particle size depends strongly on the synthesis method, part of the work is identifying which method most reliably reaches below 200 nm; candidate routes include sol–gel, sol–gel auto-combustion, hydrothermal, solvothermal, co-precipitation, spray pyrolysis, electrospinning, and microwave-assisted synthesis, selected on particle size, phase purity, and morphology.

Synthesis and validation. The top-ranked candidates will be synthesized and characterized structurally and electrochemically and the computational predictions validated against experiment.

3.3.3: Equipment Needed for Synthesis and Characterization:

The synthesis process requires various instruments for sample preparation. Below, a brief explanation of different measurement techniques and image of each piece of equipment (used in Materials Science Research Laboratory, Department of Electrical and Electronic Engineering, University of Dhaka) is provided.

Table 3.2: Measurement Techniques used in our project

Experiment / Process	Property Determined
XRD (X-ray Diffraction)	Crystal structure and phase identification
Raman Spectroscopy	Vibrational modes, bonding, and structural defects
FTIR (Fourier-transform infrared spectroscopy)	Functional groups and chemical bonding
SEM (Scanning Electron Microscopy)	Surface morphology
FESEM (Field Emission Scanning Electron Microscopy)	High-resolution surface morphology and nanostructure
TEM (Transmission Electron Microscopy)	Internal structure, particle size, and lattice fringes
XPS (X-ray Photoelectron Spectroscopy)	Surface elemental composition and oxidation states
CV (Cyclic Voltammetry)	Redox behavior and specific capacitance
GCD (Galvanostatic Charge-Discharge)	Specific capacitance, cycling stability, coulombic efficiency
EIS (Electrochemical Impedance Spectroscopy)	Charge transfer resistance and ion diffusion

BET (Brunauer-Emmett-Teller)	Surface area and pore size distribution
UV-Vis Spectroscopy	Optical absorption and bandgap
Zeta Potential	Surface charge and colloidal stability

Table 3.3: Measurement Methods targeted to be used to measure the key electrode properties

Property	Technique	How It Is Measured
Specific Capacitance	CV (Cyclic Voltammetry)	$C_{sp} = \int \frac{I dv}{2m\Delta v}$ from CV curve area
Specific Capacitance	GCD (Galvanostatic Charge-Discharge)	$C_{sp} = \frac{I\Delta t}{m\Delta V}$ from discharge slope
Surface Area	BET	N ₂ adsorption-desorption isotherm
Pore Volume	BJH / Single-point adsorption	Adsorbed volume at $P/P_0 \approx 0.99$
Pore Size Distribution	BJH / DFT model	Applied to the desorption branch
Porosity	Gas adsorption + density	From pore volume and skeletal/bulk density



(a) EKI600-2n Digital Precision Balance [22]



(b) JSHS-18D Digital Hotplate Magnetic Stirrer [21]



(c) Benchtop pH meter, CLEAN PH500 [19]



(d) Autoclave Reactor with PTFE Lined Vessel [20]



(e) Taisite WHL-25AB Desktop Drying Oven [18]



(f) Digital Hot Air Oven DSO 500D [17]



(g) Centrifuge Machine BIOBASE BKC-TH16D [24]



(h) BIOBASE MX-S Mini Vortex Shaker Lab [23]

Fig. 3.5: Various Tools Used for Weighing, Mixing, Heating, pH Monitoring, Washing, Drying in Sample Synthesis.



Fig. 3.6: Equipment for Characterization. (a) Rigaku Smartlab XRD System, Used in Centre for Advanced Research in Sciences (CARS) [25], (b) Raman Spectroscope CRS+, Used in Bangladesh Council of Scientific and Industrial Research (BCSIR) [26], (c) Shimadzu Wavelength Dispersive X-Ray-Fluorescence Spectrometer – WDXRF (XRF-1800), Used in BAEC [27], (d) JSM-7610F Schottky FESEM, Used in BCSIR [28], (e) Shimadzu UV-1900i UV-Vis-NIR Spectrometer, Used in STRC [29], (f) JED 2400 EDX Spectrometer Used in BCSIR [30], (g) Thermo Fischer Scientific K α XPS Used in BCSIR [31], (h) Thermo Talos F200X TEM, Used in BCSIR [32], (i) Shimadzu RF-6000 Spectro Fluorophotometer, Used in CARS [33].

3.4 Expected Outcome

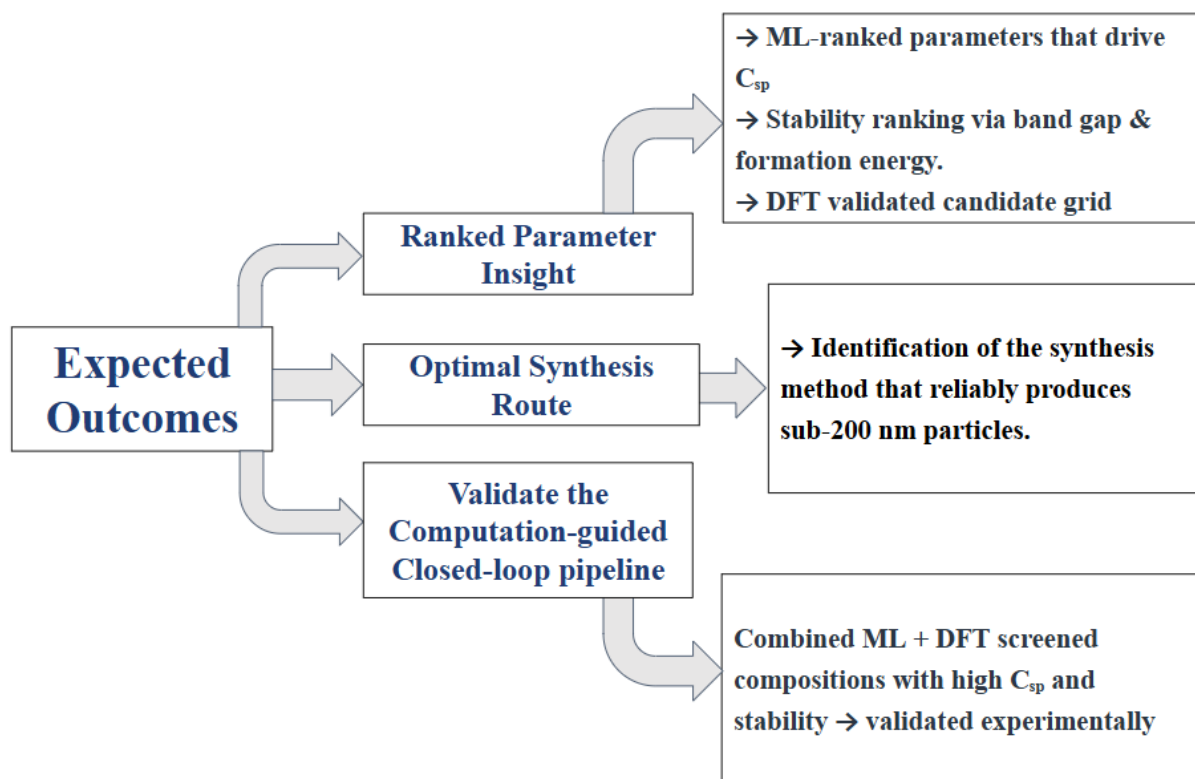


Fig. 3.7: The outline for the outcomes we expect to have reached after completion of the project.

By the end of the project, the outcomes expected to achieve are:

1. A ranked, ML-derived understanding of the parameters that strongly drive C_{sp} , plus an ML-based stability ranking (via band gap and formation energy).
2. Identification of the synthesis methods most reliably producing sub-200 nm particles of the target compositions.
3. A shortlist of compositions, found by combined ML and DFT screening and validated by synthesis, achieving as high a C_{sp} and stability as possible.

3.5 Gantt Chart

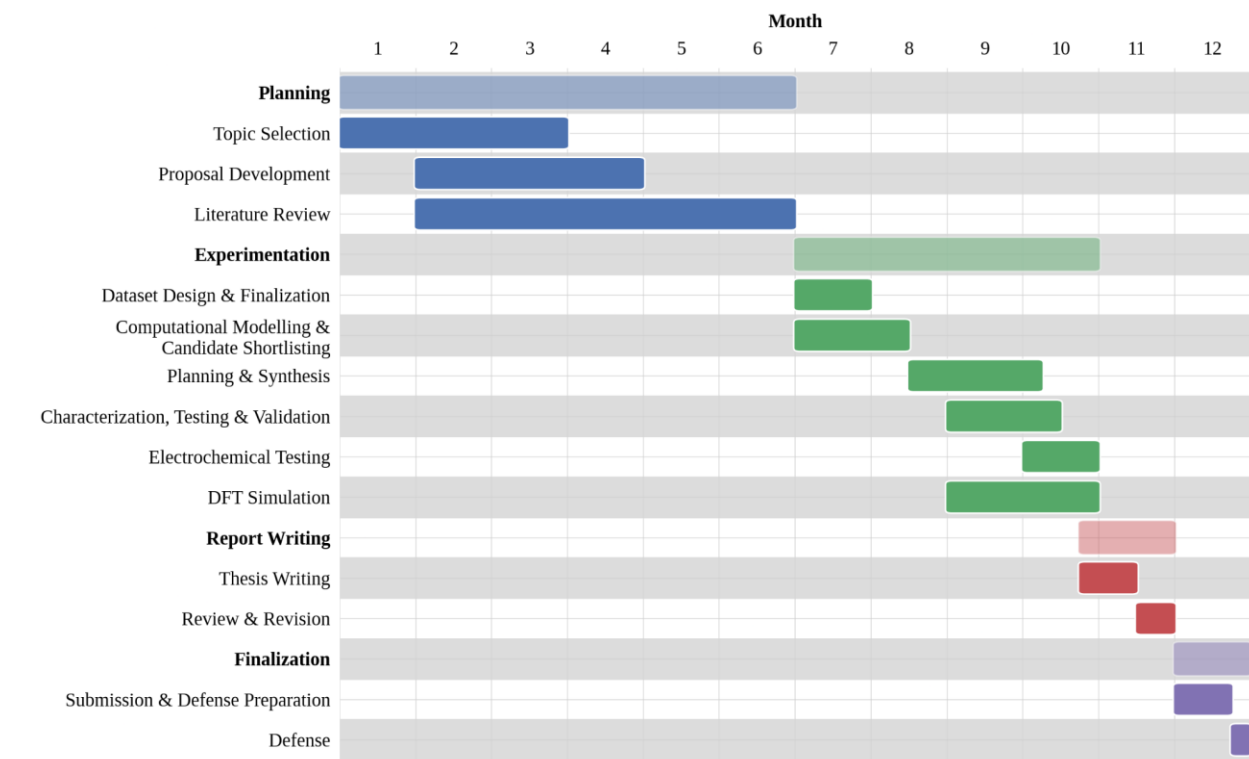


Fig. 3.8: Gantt Chart; outlining the project timeline, highlighting key phases, tasks, and milestones to keep the project on track.

CHAPTER 4

CONCLUSION

4.1 Key points of the 1st semester project:

The first semester focused on following works:

1. Identified perovskite oxides as a promising class of materials for pseudocapacitor electrodes.
2. Conducted an extensive literature-based survey of oxide supercapacitors to understand existing trends and literature gaps.
3. Compiled a descriptive dataset from literature incorporating compositional features, synthesis conditions, structural and electrochemical properties.
4. Outlined an ML based pre-screening framework to identify the best descriptors.

4.2 Future Works:

Our 2nd semester works will be following the ML-DFT-Experiment pathway. The key points of our future works are as follows:

1. Execute the ML pipeline on the constructed dataset to train and evaluate the model.
2. Perform feature-ranking analysis to identify key descriptors which influence the specific capacitance most.
3. Use ML+DFT screening to shortlist and prioritize best compositions for synthesis.
4. Carry out the synthesis of selected candidates from the shortlist.
5. Characterize the synthesized material using XRD, BET, Raman and XPS.
6. Perform electro-chemical testing of the synthesized material.
7. Validate and refine the existing ML model creating a feedback loop

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